DO NOT WRITE IN THIS AREA

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9 A, B and C are three points.

The coordinates of A are $(-2, 5)$
The coordinates of B are $(1, 1)$
The coordinates of C are $(19, -23)$

Does point C lie on the straight line that passes through A and B?
You must show how you get your answer.

Gradient AB = $\dfrac{1 - 5}{1 - (-2)} = \dfrac{-4}{3}$

1 mark
Find a gradient.

Gradient BC = $\dfrac{-23 - 1}{19 - 1} = \dfrac{-24}{18}$

1 mark
Compare another pair.

Both gradients are $-4 \div 3$.

Yes. Same gradient and common point B:

A, B and C lie on one straight line.

1 mark
Decision with working.

(Total for Question 9 is 3 marks)